Rewarding Learning

Centre Number

Candidate Number

General Certificate of Secondary Education
2026

Digital Technology

Unit 1: (Compulsory Core)
Digital Technology

[GDG11]

GDG11

MONDAY 18 MAY, AFTERNOON

TIME

1 hour.

INSTRUCTIONS TO CANDIDATES

Write your Centre Number and Candidate Number in the spaces provided at the top of this page.
You must answer the questions in the spaces provided.
Do not write outside the boxed area on each page or on blank pages.
Complete in black ink only. **Do not write with a gel pen.**
Answer **all eleven** questions.

INFORMATION FOR CANDIDATES

The total mark for this paper is 90.
Figures in brackets printed down the right-hand side of pages indicate the marks awarded to each question or part question.
Quality of written communication will be assessed in Question **8**.

For Question 1 insert the appropriate letter in the space provided.

1 **(a)** Which of the following statements about the Data Protection Act is untrue?

A Data must be processed fairly and lawfully.

B Data must be kept for longer than necessary for the purpose it was collected.

C Data must be accurate and kept up to date.

D Data must be protected against unauthorised or unlawful processing.

Answer _____ [1]

(b) Under the Consumer Contracts Regulations, how long does a consumer have to cancel an order for goods?

A 7 days

B 14 days

C 21 days

D 30 days

Answer _____ [1]

(c) Select the statement that is true about the ALU.

A The ALU stores the address of the next instruction to be fetched.

B The ALU stores all the programs and data, temporarily, whilst they are in use.

C The ALU decodes instructions so that they can be executed.

D The ALU performs logical comparisons to assist with decision making.

Answer _____ [1]

(d) What is a computer called when it is not connected to a network?

A A Router

B A Switch

C A Standalone

D A Wi-Fi

Answer _____ [1]

(e) Select the statement which best describes data compression.

A Alphabetically organising data to improve readability.

B Making a file smaller by removing unnecessary data.

C Changing data into a different format for easier editing.

D Making a file larger for better quality and performance.

Answer _____ [1]

(f) Which of the following best describes cloud computing?

A Storing and processing data on servers accessed through the Internet.

B Running all programs only on your computer without Internet access.

C Saving data on hard drives connected directly to your computer.

D Using one computer to do all tasks without any network connection.

Answer _____ [1]

[Turn over

(g) What does the term buffering mean?

A Temporarily storing data to manage speed differences between devices.

B Compressing data to make files smaller.

C Encrypting sensitive data.

D Transferring data directly between devices.

Answer _____ [1]

(h) Select the statement which best describes a pixel.

A The smallest part of a digital image.

B A graphic that can be resized without losing quality.

C A format for saving vector images.

D A way to create animations in digital media.

Answer _____ [1]

BLANK PAGE

DO NOT WRITE ON THIS PAGE

(Questions continue overleaf)

[Turn over



2 **(a)** Describe how bit rate affects the quality of sound when recording.

[2]

(b) Why is analogue-to-digital conversion necessary in sound recording?

[3]

3 What do the letters below stand for?

(a) PDF

_____ [1]

(b) RTF

_____ [1]

[Turn over

4 Academy Sports Club uses a database to store data about their members and bookings. The tables below contain sample data from Academy Sports Club.

tblMEMBER

MemberID	Name	DOB	Membership	DateJoined
M001	John Smith	12/05/1990	Premium	01/01/2023
M002	Sarah Jones	23/08/1985	Basic	15/02/2023
M003	Emily Brown	02/11/1995	Premium	10/01/2023

tblACTIVITY

ActivityID	Activity	StartTime	EndTime	Price
A001	Yoga	10:30	12:00	£35.00
A002	Weightlifting	18:00	19:30	£45.00
A003	Pilates	09:00	10:00	£25.00

tblBOOKINGS

BookingID	MemberID	ActivityID	Date
B001	M001	A001	15/03/2026
B002	M002	A002	16/03/2026
B003	M001	A003	17/03/2026

(a) State the name of the field in **tblMEMBER** that is used as a primary key.

_____ [1]

The table below contains several fields from the database.

(b) Choose the correct data type for each one.

Field Name	Data Type
ActivityID	
DOB	
Price	

[3]

(c) Circle the term which describes the relationship between **tblMEMBER** and **tblBOOKINGS**.

One to One **One to Many** **Many to Many** [1]

Academy Sports Club wants to retrieve all bookings for member John Smith.

(d) What feature in a database program would they use?

(i) To retrieve this information.

_____ [1]

(ii) To present the information retrieved.

_____ [1]

[Turn over

(e) Organisations such as Academy Sports Club may make use of Big Data.

(i) Explain what is meant by the term Big Data.

_____

_____ [2]

(ii) What is data analytics?

_____

_____ [2]

BLANK PAGE

DO NOT WRITE ON THIS PAGE

(Questions continue overleaf)

[Turn over



5 **(a)** Complete the paragraph below using the words provided about computers.

Communicate **Resources** **Cables** **Network**

A _____ [1] is a group of computers linked together.

They are connected usually by _____ [1] or Wi-Fi.

Computers connected together can share _____ [1],

such as files and printers, and are able to _____ [1]

with each other.

(b) What is the name given to a powerful computer that acts as a central hub in a network?

_____ [1]

A protocol is a set of rules/standards that govern the transmission of data over a network.

(c) State **two** protocols used to transmit data.

1. _____

2. _____ [2]



(d) The table below contains statements about a ring network.

Place a tick (✓) beside **three** statements that are true about a ring network.

Statement	**Tick (✓)**
Data travels in one direction.	
The main cable has terminators.	
Data collisions frequently occur in a ring network.	
Adding a workstation will disrupt the network.	
All workstations have equal access.	

[3]

[Turn over

6 Stationery Supplies uses a spreadsheet to track their monthly sales. Below is an example of the SALES figures.

	A	B	C	D	E	F	G	H
1	**Stationery Supplies**				**May**	**2026**		
2								
3	**OrderID**	**Customer Name**	**Product**	**Order Date**	**Qty Sold**	**Unit Price**	**Total Sales**	**Discount**
4	101	Logan Brothers	Notebook	15/05/2026	10	£1.99	£19.90	5%
5	102	Prenton Clothes	T-shirt	15/05/2026	15	£2.99	£44.85	10%
6	107	Brown Brothers	Mug	17/05/2026	20	£3.99	£79.80	12%
7	108	Cards Galore	Diary	20/05/2026	15	£5.99	£89.85	12%
8	109	Abbey School	Key rings	25/05/2026	100	£1.25	£125.00	15%
9								
10				Mean Sales for May			£71.88	

(a) Complete the sentence below using the words provided.
(Not all words will be used.)

Relative **Specific** **Absolute** **Formula**

An _____ [1] cell reference in a spreadsheet stays fixed on a _____ [1] cell, even when the _____ [1] is copied to other cells.

(b) What feature in a spreadsheet can Stationery Supplies use to ensure that the Qty Sold column contains only numbers?

_____ [1]

(c) What is the formula contained in cell G4?

_____ [1]

(d) What is the term used to describe copying a formula or data from one cell to another in a spreadsheet?

_____ [1]

(e) What is the formula used in cell G10 to calculate the Mean Sales for May?

_____ [1]

[Turn over

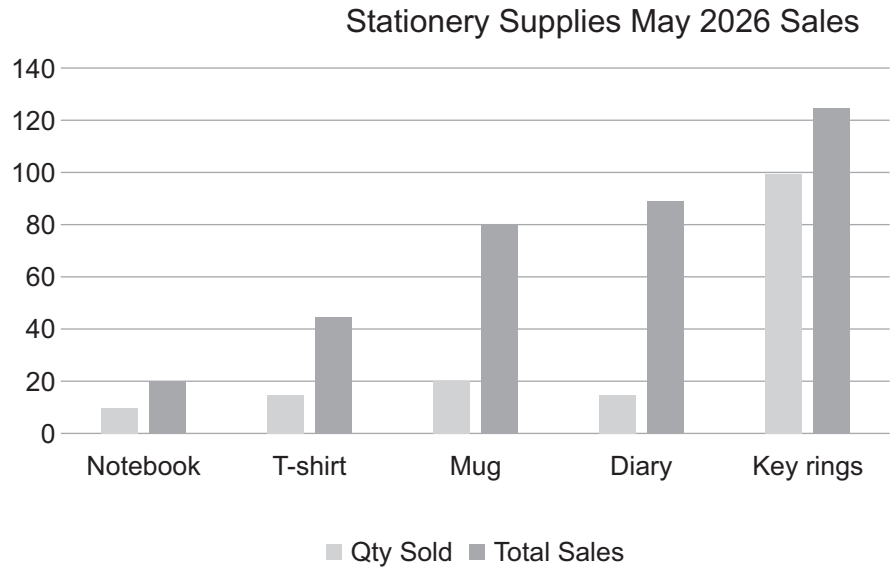
(f) Stationery Supplies uses a spreadsheet for data modelling. What is data modelling?

_____

_____ [2]

Stationery Supplies produce a graph showing the sales figures for May 2026.

(g) State the cell ranges used to produce the graph below.



1. _____

2. _____

3. _____ [3]

7 Riek is buying a new computer and is confused by some of the terminology the sales adviser is using.

(a) Complete the table below by placing a tick (✓) beside **three** facts that are true about the Central Processing Unit (CPU).

Facts	**Tick (✓)**
The CPU performs calculations.	
The CPU stores large amounts of data permanently.	
The CPU can run software without loading it into memory.	
The CPU processes information provided by input devices.	
The CPU runs the operating system and application software.	

[3]

The sales adviser informs Riek that she needs a computer with a clock speed between 2.5 GHz and 4.0 GHz.

(b) What is the benefit of a computer having a high clock speed?

_____

_____

_____ [2]

Computers have different types of memory.

(c) Which type of memory:

(i) Is non-volatile and permanently stores data?

_____ [1]

(ii) Loses its contents when the computer is turned off?

_____ [1]

[Turn over

8 E-Commerce has impacted everyday living.

Describe the advantages of e-Commerce under the following two headings.

Quality of written communication will be assessed in this question.

1. Changes in shopping behaviour

2. Customer interaction

[6]

9 Employers have responsibilities to their employees regarding their working environment.

(a) In the table below, place a tick (✓) beside **three** responsibilities an employer has to their employee regarding health and safety.

Responsibility	**Tick (✓)**
Provide suitable lighting	
Ignore trailing wires	
Provide ergonomically designed furniture	
Make employees pay for regular eye tests	
Keep equipment in good repair	

[3]

Taking regular breaks during work helps prevent health issues.

(b) List **two** health issues that can be avoided by taking regular breaks.

1. _____

2. _____ [2]

Extended computer use can lead to health problems.

(c) State **three** elements that a well designed workstation should include.

1. _____

2. _____

3. _____ [3]

[Turn over

10 Place the correct term below against the appropriate definition.

Spyware **Velocity** **Hacking** **Switch** **CSV**

Simulation **Register** **Bluetooth** **Cache** **Laser Printer**

Definition	Term
Manages communication between the file server and network computers	
The speed at which data is created, handled and examined	
A location in memory where data or information is stored temporarily	
Unauthorised access to computer systems	
Software that lets a user monitor another person's computer activity	
Stores frequently accessed programs or instructions	
A file format where data is separated by commas	
Produces high quality printouts very quickly	
A computer program that predicts how a real-life system might behave	
Communicates using radio frequencies	

[10]

11 Processing is carried out by a computer system.

(a) Draw a line from the statement and match it with the correct processing method.

Statement	Processing Method
Used in billing systems	Real Time
Used in life critical systems, e.g. air traffic control	Real Time
Data is processed instantly as it arrives	Batch
Data is processed and completed later	Batch

[4]

(b) List **two** resources managed by the operating system.

1. _____

2. _____ [2]

(c) Complete the paragraph below using the words provided.

Defragmenter **Manage** **Virus**

Utility programs help to maintain and _____ [1] the computer system. An example of a utility program is a _____ [1] scanner, which helps to protect against malware. A disk _____ [1] is a utility program that optimises how data is stored on a hard drive.

[Turn over

Making regular backups of the data stored on a computer is crucial.

(d) State **two** reasons for backing up computer data.

1. _____

_____

2. _____

_____ [2]

THIS IS THE END OF THE QUESTION PAPER

BLANK PAGE

DO NOT WRITE ON THIS PAGE

DO NOT WRITE ON THIS PAGE

For Examiner's use only	
Question Number	**Marks**
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
Total Marks	

Examiner Number

Permission to reproduce all copyright material has been applied for.
In some cases, efforts to contact copyright holders may have been unsuccessful and CCEA will be happy to rectify any omissions of acknowledgement in future if notified.